

GENERIC REFINERY - GULF COAST

100 kBPSD Crude & Vacuum Distillation Unit

Process Design Report - Simulation & Equipment Sizing

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1 Purpose

This report documents the process simulation and equipment sizing basis for the 100 kBPSD Crude & Vacuum Distillation Unit (FEED, Rev A). It summarises the method, the heat and material balance (full stream table in CFU-000-PR-HMB-001) and the hydraulic and thermal sizing of the main equipment. Every number is produced by the calculation code in this repository (python build.py). That code is the calculation record.

2 Method

2.1 Crude characterisation

The design crude TBP curve is split into 37 pseudo-components (15 C cuts to 395 C, 25 C to 595 C, then 50 C), plus discrete light ends C2-nC4. Specific gravity uses a Watson K that varies with boiling point, $K = K_0 - 0.00025(T_b - 100)$, with $K_0 = 11.927$ fitted to the bulk 33.4 API. MW, T_c and P_c come from Riazi-Daubert (1987) and the acentric factor from Lee-Kesler. Vapour pressure uses Lee-Kesler corresponding states. Enthalpy: Watson-Nelson liquid C_p , Fallon-Watson vapour C_p , Kistiakowsky latent heat at T_b (reference liquid at 15 C). Sulfur is distributed by boiling point and scaled to 1.8 wt%.

2.2 Column modelling (cut-point model)

- Products are defined by TBP cut points with a sigmoid overlap ("sloppy split"); the width increases down the column, matching typical crude-unit ASTM gaps and overlaps.
- Flash zones: equilibrium (Raoult) flash with stripping steam as an inert. The flash-zone temperature is solved so that vapour = all distillates + overflash (3% LV on crude; vacuum 3% LV on feed). $COT = FZ + \text{transfer-line temperature drop}$.
- Draw temperatures are the product bubble point at the tray's hydrocarbon partial pressure; steam and internal reflux are iterated with the heat balance.
- Total heat removal comes from the overall column enthalpy balance. It is split between top reflux and pumparounds (TPA 17%, MPA 27%, BPA 26%); internal reflux below each draw comes from envelope balances, and gives the section vapour and liquid loads used for tray sizing.
- Stabiliser and splitter: Fenske-Underwood-Gilliland short-cut, Fenske distribution of non-keys, Kirkbride feed stage, $R = 1.3 R_{min}$, tray efficiency 75 %.
- Preheat train: sequential counter-current exchangers, minimum approach 20 C; the remaining duty goes to trim coolers or steam generators.

2.3 Limitations (FEED accuracy)

The thermodynamics are ideal (Raoult), and the model does not do tray-to-tray rigorous rating. Expect the flash-zone and draw temperatures to be within about 5-10 C of a rigorous simulation (HYSYS/Petro-SIM). Before detailed engineering, these numbers must be confirmed with a rigorous simulation on a full crude assay.

Model warnings:

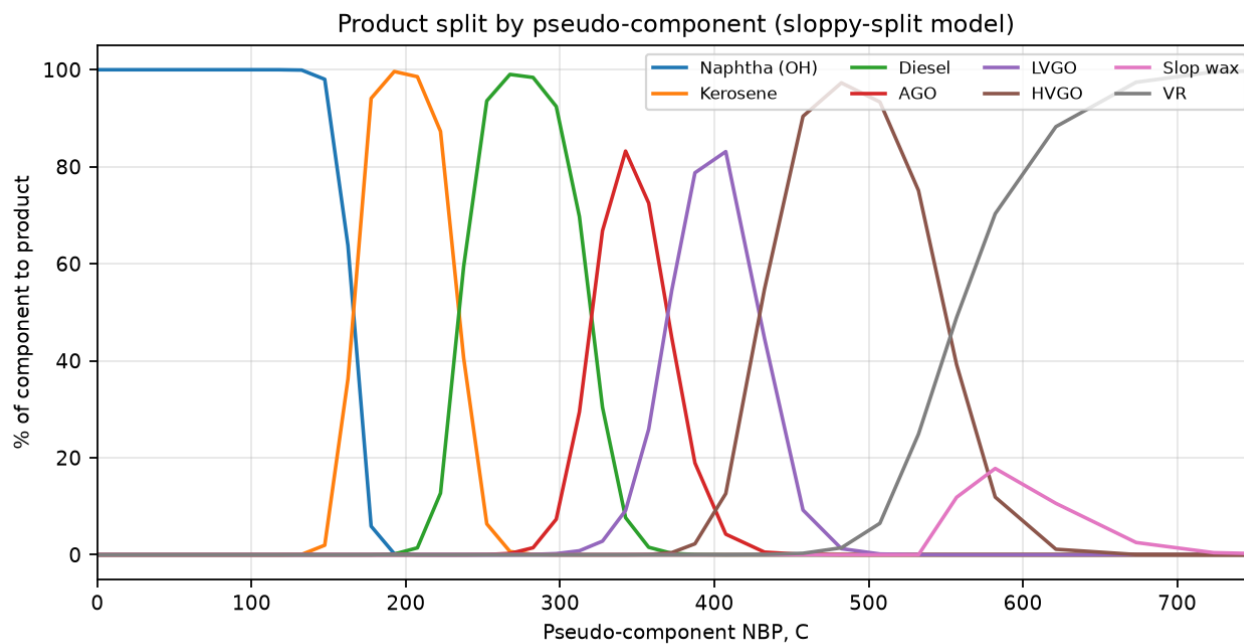
- None

3 Heat and material balance summary

3.1 Product yields

Product	Stream	BPSD	t/h	wt%	API	TBP5 C	TBP95 C	S wt%
LPG	30	1,743	6.5	1.14	120.8	-76	39	0.001
Light naphtha	32	5,920	27.6	4.86	69.3	13	84	0.018
Heavy naphtha	33	16,275	80.8	14.23	57.0	83	167	0.055
Kerosene	16	13,017	68.6	12.08	46.2	163	239	0.238
Diesel	17	14,701	81.6	14.37	37.2	230	326	0.948
AGO	18	7,930	45.8	8.07	30.5	306	387	1.779
LVGO	23	8,467	50.4	8.87	25.9	350	451	2.295
HVGO	24	14,553	90.4	15.92	19.3	418	560	2.892
Slop wax	25	1,212	7.9	1.39	12.3	542	672	3.751
Vacuum residue	26	16,181	108.3	19.06	8.5	537	824	4.022

Overall hydrocarbon balance: products + off-gases = 567.87 t/h vs crude 567.87 t/h (+0.000 %).



Product split by pseudo-component

3.2 Fractionation quality (ASTM D86 estimated via Riazi-Daubert TBP-D86)

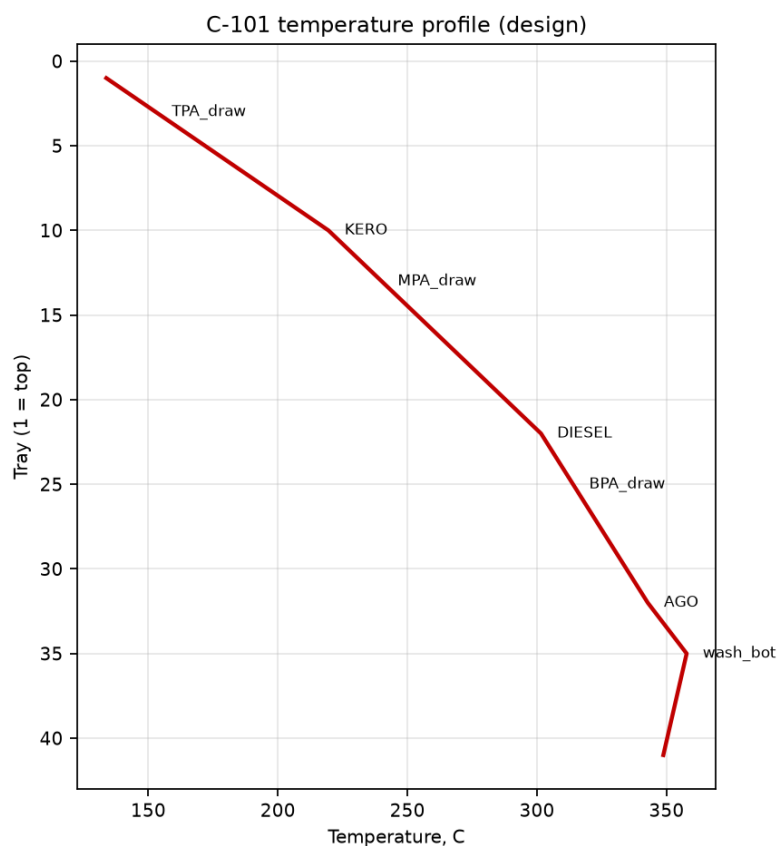
Pair	Light D86 90%	Heavy D86 10%	Light D86 EP	Heavy D86 IBP	5-95 gap (IBP-EP), C
Heavy naphtha / Kerosene	150	184	172	181	+9
Kerosene / Diesel	221	251	248	248	+1
Diesel / AGO	303	322	335	327	-8

A negative IBP-EP gap means overlap; typical crude-unit targets are a gap of +10 C (naphtha/kero) and an overlap of 0 to -20 C (kero/diesel, diesel/AGO).

4 Atmospheric section

Parameter	Value
Crude inlet temperature (CIT)	274.0 C
Desalter temperature	136.4 C
H-101 COT	365.7 C
Flash zone T / P	360.7 C / 1.47 barg
Column top T / P	133.8 C / 1.19 barg
Top water dew point (margin)	83.0 C (51 C)
Bottom T	348.7 C
Reflux (to top tray)	101.3 t/h (R/D = 0.88)
Draw temperatures K / D / AGO	220 / 302 / 343 C
Total heat removal	47.58 MW
Overhead condensing duty	39.70 MW
Stripping steam bottom/K/D/AGO, kg/h	7638 / 1476 / 1667 / 899

Pumparound	Draw tray	Return tray	Duty MW	Draw T C	Return T C	Circulation t/h
TPA	3	1	8.09	153	83	167
MPA	13	11	12.85	240	160	208
BPA	25	23	12.37	314	224	166



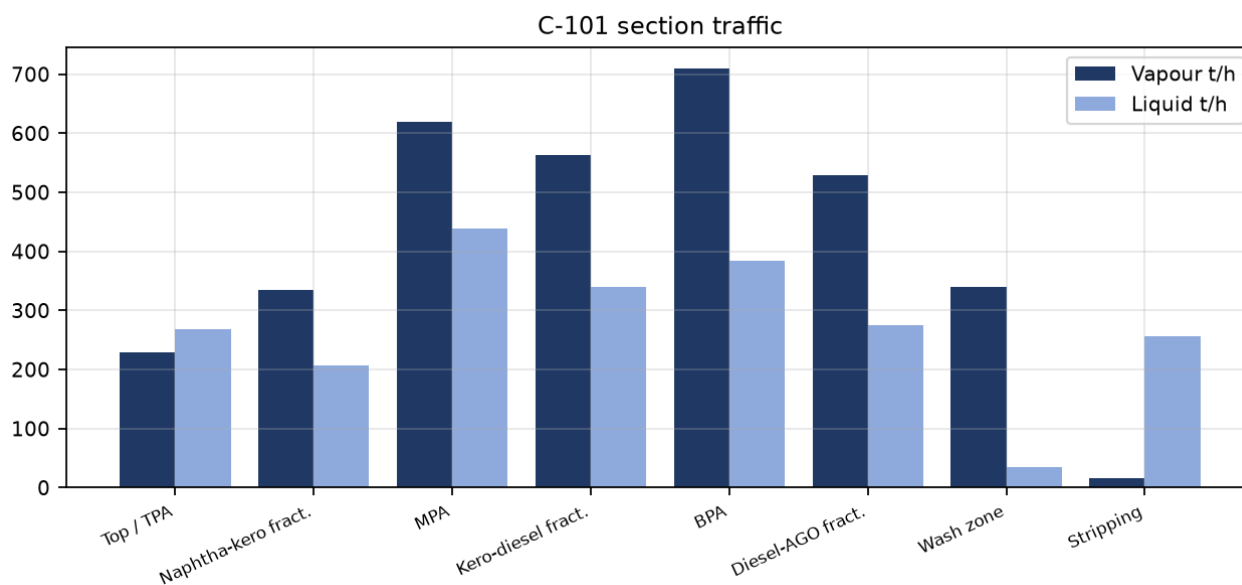
C-101 temperature profile

4.1 C-101 tray hydraulics (Fair flooding, 80 % flood, 85 % net area)

Section	Trays	V t/h	L t/h	T C	rho V	rho L	FLV	TS mm	u flood m/s	D calc m
Top / TPA	1-3	229	268	139	5.02	611	0.106	760	1.12	4.60
Naphtha-k ero fract.	4-9	335	207	181	5.61	651	0.057	610	1.02	5.51
MPA	11-13	620	438	233	7.53	599	0.079	760	0.94	6.74

Section	Trays	V t/h	L t/h	T C	rho V	rho L	FLV	TS mm	u flood m/s	D calc m
Kero-diesel fract.	14-21	564	339	267	8.08	621	0.069	610	0.81	6.68
BPA	23-25	710	384	310	9.24	578	0.068	760	0.85	6.85
Diesel-AGO fract.	26-31	529	275	326	9.49	610	0.065	610	0.75	6.23
Wash zone	33-35	339	34	358	6.03	635	0.010	610	1.07	5.24
Stripping	36-41	15	257	354	1.60	706	0.796	610	0.82	2.47

Selected: ID 6.9 m (top/main) / 4.1 m (stripping) x 36.0 m T/T. Internals: 41 valve trays (2/4-pass), 410S; Monel-lined top 5 trays.



C-101 section traffic

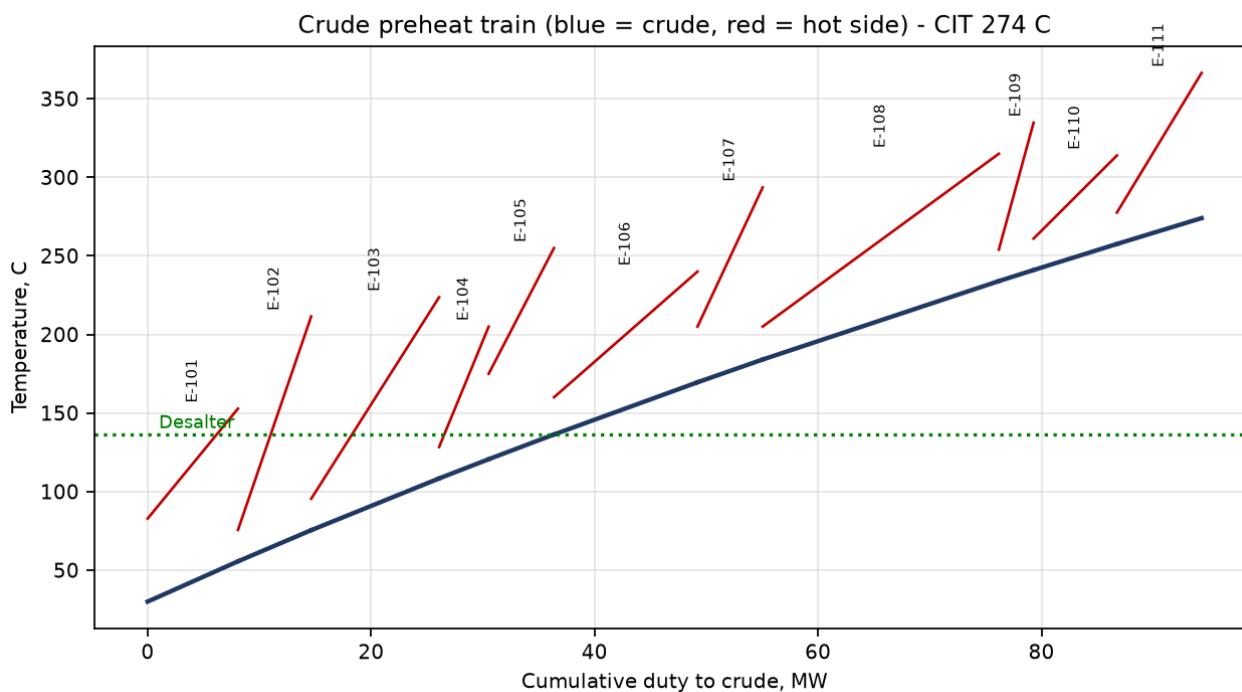
5 Crude preheat train

Exchanger	Hot stream	Duty MW	Hot in/out C	Crude in/out C
E-101	TPA	8.09	153 / 83	30 / 56
E-102	KERO	6.55	212 / 76	56 / 75
E-103	LVGO	11.45	224 / 95	75 / 108
E-104	DIESEL	4.44	205 / 128	108 / 121
E-105	VR	5.84	255 / 175	121 / 136
E-106	MPA	12.85	240 / 160	136 / 170
E-107	DIESEL	5.83	294 / 205	170 / 184
E-108	HVGO	21.14	315 / 205	184 / 234
E-109	AGO	3.11	335 / 254	234 / 241
E-110	BPA	7.46	314 / 261	241 / 258
E-111	VR	7.58	366 / 278	258 / 274

Exchanger	Area m2	U W/m2K	F	Shells (series x parallel)
E-101	359	330	0.94	1 x 1
E-102	438	300	0.82	1 x 1
E-103	804	280	0.93	2 x 1
E-104	378	290	0.91	1 x 1
E-105	431	170	0.97	1 x 1
E-106	1001	320	0.94	2 x 1
E-107	324	290	0.94	1 x 1
E-108	2196	250	0.87	2 x 2
E-109	263	260	0.95	1 x 1
E-110	886	280	0.86	1 x 2
E-111	1094	170	0.86	1 x 2

Trim duties (heat not recovered to crude):

Tag	Stream	Type	Duty MW	T C
E-113	BPA	MP steam generator	4.91	261 -> 224
A-103	KERO	air cooler	1.27	76 -> 45
A-104	DIESEL	air cooler	3.72	128 -> 55
A-105	AGO	air cooler	6.06	254 -> 60
A-201	LVGO	air cooler	3.10	95 -> 55
E-201	VR	LP steam generator	1.78	278 -> 255
A-202	HVGO product	air cooler	6.64	205 -> 90



Preheat train temperature-duty diagram

6 Fired heaters

Item	H-101	H-201
Process duty, MW	59.55	14.69
Absorbed duty, MW	60.68	14.69
Fired duty (LHV), MW	67.42	16.7
Efficiency	0.9	0.88
Inlet T, C	274.0	349.0
Outlet T (COT), C	365.7	398.4
Outlet P, bar(a)	3.38	0.31
Outlet vapour fraction (mass)	0.533	0.264
Radiant duty, MW	38.71	9.55
Radiant area, m2	1229.0	382.0
Radiant tubes (6 in x 18.3 m)	128	40
Passes	8	4
Mass flux, kg/m2s	1176.0	1066.0
Burners	16	6
Fuel gas, kg/h	5164.0	1279.0

7 Vacuum section

Parameter	Value
Feed (atmospheric residue)	256.9 t/h, 268 m3/h
H-201 COT	398.4 C
Flash zone T / P	386.4 C / 60 mbar(a)
Top T / P	70 C / 20 mbar(a)
LVGO / HVGO draw T	224 / 315 C
Bottom T (quenched)	366 C

Parameter	Value
Stripping + coil steam	2789 kg/h
Total heat removal	22.59 MW (LVGO PA 9.0, HVGO PA 13.6)
Ejector motive steam	9352 kg/h MP steam

C-201 bed sizing (packing at $C_s = 0.11$ m/s, wash grid 0.12 m/s; stripping trays at 75 % flood):

Section	V t/h	L t/h	T C	P mbar	rho V	Q m3/s	D calc m
Bed 1 - LVGO PA	10.4	83	147	22	0.037	79	2.49
Bed 2 - LVGO/HVGO fract.	87.2	34	285	34	0.176	138	4.97
Bed 3 - HVGO PA	177.6	161	347	44	0.283	174	6.41
Bed 4 - Wash	151.9	13	382	56	0.307	137	5.50
Flash zone	151.9	108	386	60	0.327	129	5.60
Stripping	5.0	108	371	70	0.041	35	2.49

Selected: ID 2.6 m (top) / 6.6 m (main) / 3.0 m (boot) x 36.0 m T/T.

8 Light ends

Item	C-105 stabiliser	C-106 splitter
Light / heavy key	nC4 / iC5-range pseudo	NBP<80 / NBP>80 pseudo
Top / bottom P, bar(a)	11.80 / 12.30	2.10 / 2.50
Top / bottom T, C	79 / 203	84 / 152
Nmin / Rmin / R	7.2 / 1.45 / 1.89	14.6 / 1.60 / 2.08
Theoretical / actual trays	15.2 / 19	29.5 / 38
Feed tray (from top)	13	21
Condenser / reboiler duty, MW	2.14 / 10.83	9.63 / 6.83
Diameter x height	ID 2.1 m x 18.0 m T/T	ID 3.1 m x 29.5 m T/T

9 Pumps

Tag	Service	Rated m3/h	Head m	Abs. kW	Motor kW	API 610
P-101A/B	Crude charge	730	261	617	710	BB2
P-102A/B	Desalted crude booster	805	263	616	710	BB2
P-103A/B	Atm. reflux	159	102	46	55	OH2
P-104A/B	Unstabilised naphtha	180	232	119	160	OH2
P-105A/B	Overhead sour water	14	51	3	5.5	OH2
P-106A/B	TPA pumparound	293	130	94	110	OH2
P-107A/B	MPA pumparound	387	138	123	160	BB2
P-108A/B	BPA pumparound	318	142	102	132	BB2
P-109A/B	Kero product	122	148	46	55	BB2
P-110A/B	Diesel product	151	154	56	75	BB2
P-111A/B	Ago product	84	153	32	45	BB2
P-112A/B	Atm. residue / vacuum heater charge	398	172	189	250	BB2

Tag	Service	Rated m3/h	Head m	Abs. kW	Motor kW	API 610
P-114A/B	Desalter wash water	37	82	13	18.5	OH2
P-115A/B	Stabiliser reflux / LPG	39	154	14	18.5	OH2
P-116A/B	Splitter reflux / LN	146	111	43	55	OH2
P-117A/B	Heavy naphtha product	142	98	36	45	OH2
P-118	Desalter mud-wash / recycle	19	64	5	7.5	OH2
P-201A/B	LVGO pumparound / product	198	138	81	110	BB2
P-202A/B	HVGO pumparound / product	391	144	155	200	BB2
P-203A/B	Slop wax	12	114	5	5.5	BB2
P-204A/B	Vacuum residue (incl. quench)	196	187	112	132	BB2
P-205A/B	Hotwell sour water	13	51	3	3.7	OH2
P-206A/B	Hotwell slop oil	1	60	0	0.75	OH2

10 Relief loads (preliminary governing cases)

Tag	Protects	Case	Load kg/h	Set barg	Orifice
PSV-1001	C-101	Reflux failure / blocked OH (total OH vapour)	216,219	2.4	4 x T
PSV-1002	D-101A	Fire (liquid full)	50,292	15.5	1 x M
PSV-1003	D-101B	Fire (liquid full)	50,292	15.5	1 x M
PSV-1004	D-102	Fire	12,890	2.4	1 x P
PSV-1005	C-105	Reflux failure (reboiler duty)	129,912	13.0	1 x R
PSV-1006	D-105	Fire	5,902	15.2	1 x H
PSV-1007	C-106	Reflux failure (reboiler duty)	76,867	3.5	1 x T
PSV-1008	E-116	Tube rupture (HP steam into naphtha)	25,000	13.0	1 x P
PSV-1009	C-102/103/104	Fire (side strippers)	15,948	3.5	1 x N
PSV-1010	P-101 disch.	Blocked outlet / thermal	5,000	30.0	1 x E
PSV-2001	C-201	Loss of vacuum / fire (blocked outlet to ejectors)	32,789	3.5	1 x R
PSV-2002	D-201	Fire	6,324	3.5	1 x K

The global flare load (power failure, cooling failure) is assessed in the flare study. The largest single unit contributor is the C-101 overhead (reflux failure).